



ThermalODE: A Physics-Informed Simulator for Multi-GPU HPC Clusters



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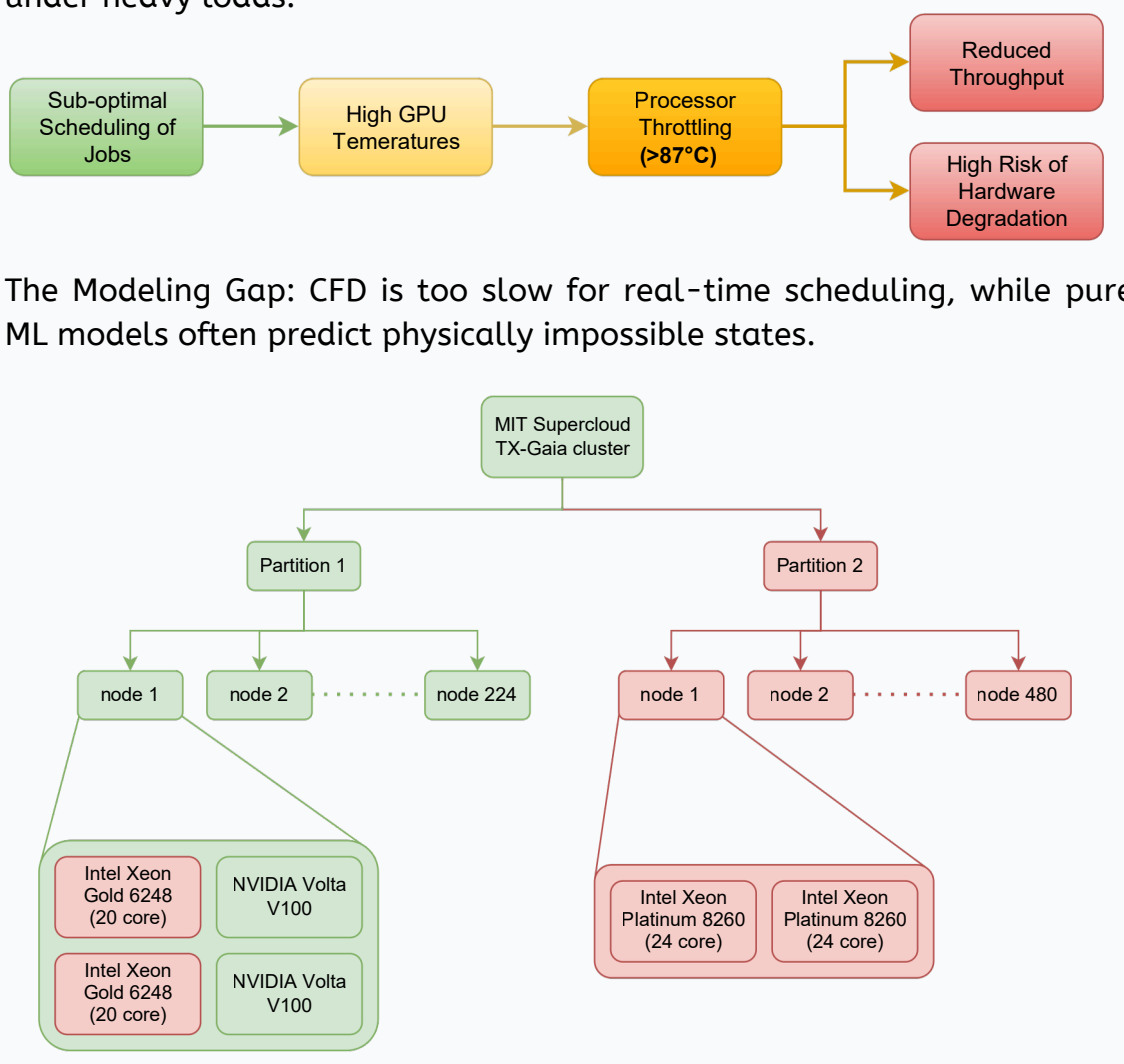
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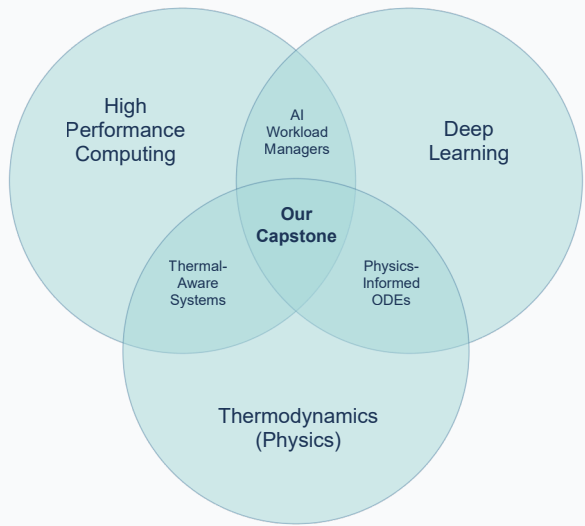
Problem Statement

- The Problem: Traditional HPC schedulers (like Slurm) ignore real-time thermals, causing high-density GPU clusters to throttle and lose efficiency under heavy loads.
- The Modeling Gap: CFD is too slow for real-time scheduling, while pure ML models often predict physically impossible states.



Innovation

Our innovation doesn't stop at the intersection of High Performance Computing, Deep Learning and Physics. By prioritizing thermal efficiency, ThermalODE directly contributes to the United Nations Sustainable Development Goals, proving that advanced computing can drive global sustainability.



Methodology

Phase 1: Dual-GPU Validation & Cleaning

1

Type Safety & Parquet

22,471

TOTAL FILES PROCESSED

Raw CSVs were converted to PyArrow Parquet format for strict schema enforcement, massive compression, and rapid I/O operations.

2

Dual-GPU Consistency

6,681

VALID FILES RETAINED

Filtered out nodes that lacked perfectly overlapping GPU 0 and GPU 1 telemetry traces (max 5% row count diff, 5s overlap tolerance).

3

Thermodynamic Activity

6,629

FINAL CLEANED FILES

Removed files lacking dynamic thermal excitation (required at least a 1.0°C or 5.0W delta). Timestamps normalized to t=0.

Phase 2: 2D Stratified Split

THE 2D STRATIFICATION MATRIX

len_bin \ den_bin	Low	Medium	High	Total
Short	938	1028	244	2210
Medium	401	715	1093	2209
Long	871	466	873	2210
Total	2210	2209	2210	6629

Phase 3: ODE Tensor Chunking

Jobs were segmented into exactly 5,000-row arrays (padding shorter sequences). To prevent RAM exhaustion during PyTorch data loading, training segments were stacked into massive .pt chunk files, while test files were mapped 1-to-1 to preserve true physical trajectories.

TRAIN SEGMENTS

343,503

Batched in .pt chunks

VAL SEGMENTS

39,533

Combined 1 Bundle

TEST SEGMENTS

44,691

Mapped 1:1 Files

Phase 4: Thermodynamic Prior Calculations

Empirical Physics Initialization

Gradient descent performs best when initialized near the true global minima. Specialized scripts were written to scan the dataset for isolated asymmetric heating events and to fit exponential step-response curves to the data. This allows the calculation of the real-world thermal resistances and crosstalk boundaries of the hardware before ODE training began.

Technical Implementation

DIE TEMPERATURE DERIVATIVE

$$\frac{dT_{die}}{dt} = \frac{P_{self} - \frac{T_{die} - T_{sink}}{R_{paste}}}{C_{die}}$$

SINK TEMPERATURE DERIVATIVE

$$\frac{dT_{sink}}{dt} = \frac{T_{die} - T_{sink}}{R_{paste}} + k \cdot P_{adj} - h_{fan}(T_{sink} - T_{amb}) + q$$

FORWARD EULER INTEGRATION

$$T_{die}(t + \Delta t) = T_{die}(t) + \Delta t \cdot \frac{dT_{die}}{dt}$$
$$T_{sink}(t + \Delta t) = T_{sink}(t) + \Delta t \cdot \frac{dT_{sink}}{dt}$$

PARAMETER BOUNDS

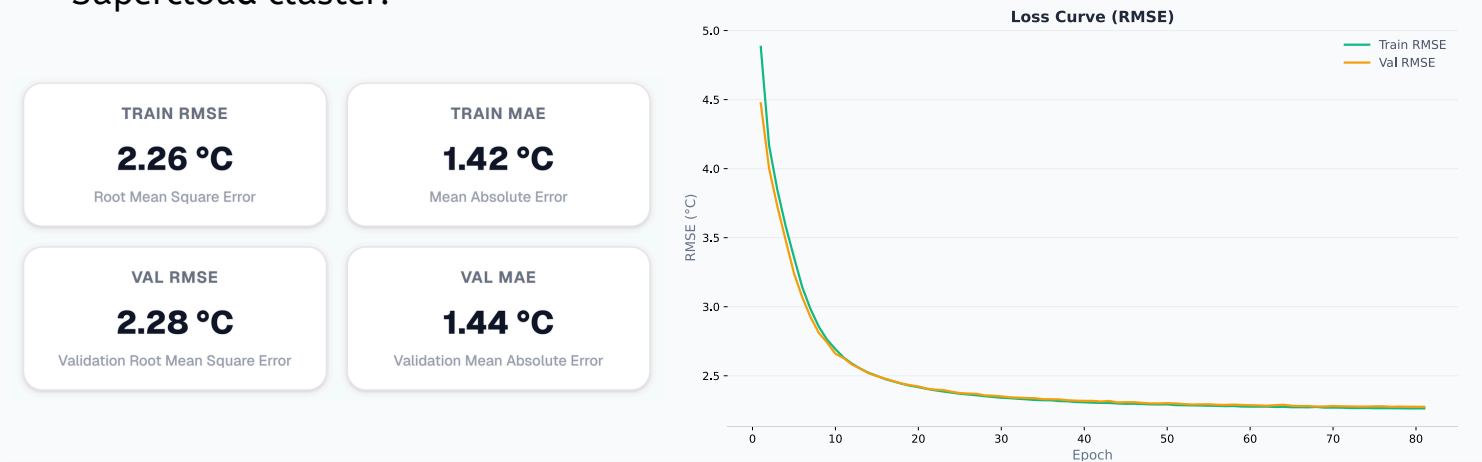
$$norm = \frac{val - low}{high - low}$$
$$\sigma^{-1}(norm) = \ln\left(\frac{norm}{1 - norm}\right)$$

NON-LINEAR FAN CURVE

$$h_{fan} = h_{base} + h_{active} \cdot \sigma(\beta(T_{die} - T_{thresh}))$$

Results & Validation

- The model was evaluated on over 443 million unseen timesteps across 671 jobs, and achieved impressive results, validating its use as a digital twin for the MIT Supercloud cluster.



Final Model Testing Metrics (Inference)

443,815,380 Timesteps

GLOBAL ACCURACY	TAIL-RISK METRICS	STABILITY ZONES
RMSE 2.2849 °C	95th Percentile Err 4.6601 °C	Danger Zone RMSE 5.3853 °C
MAE 1.4561 °C	99th Percentile Err 7.9498 °C	First 10% RMSE 3.7968 °C
Directional Bias +0.1955 °C	Absolute Max Err 34.6839 °C	Last 10% RMSE 2.1676 °C